

# Alexandre AUTRAN

Mathematics and Machine Learning Student

📍 Rennes, France

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## Profile

I am a mathematics and machine learning student (*élève normalien*) at ENS Rennes and ENS Paris-Saclay. My research interests include high-dimensional statistics, stochastic processes, optimal transport, and optimal control. I combine theoretical analysis with numerical methods and machine learning implementations in Python, with a current interest in graph learning and generative models.

## Education

### 2024–2027 Normalien – ENS Diploma in Mathematics

Admitted to the ENS through a competitive entrance examination.  
Four-year diploma pursued over three years (accelerated track).

2026–2027 **Research Year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, France  
Final year of the ENS Diploma (research track, “Prelab” ENS).

2025–2027 **MSc in Mathematics**, *École Normale Supérieure de Rennes*, France  
M2 Fundamental Mathematics. Highest honors (17/20 in the first semester, completed alongside M1).  
M1 Fundamental Mathematics. Highest honors (17.6/20). Rank: 5/100+.

2024–2025 **BSc in Mathematics, minor in Computer Science**, *École Normale Supérieure de Rennes*, France  
L3 Fundamental Mathematics. High honors (15.4/20).

2022–2024 **BSc in Mathematics, minor in Computer Science**, *University of Rennes*, France  
L2 Mathematics. Highest honors (18/20). Rank: 3/100+.  
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

## Publications

[\[More details\]](#)

- [1] **A. Autran**, Conditional Fluctuations of Mean-Field Systems with Stable Common Noise, preprint, 2026.  
Under review at *Electronic Journal of Probability*.
- [2] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, preprint, 2026.  
Submitted to *Stochastic Processes and their Applications*.
- [3] **A. Autran** et al., Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices  
Subjected to Single Ionizing Particles, *Applied Sciences*, 16(11), 5551, 2026.

## Research Experience

[\[More details\]](#)

### Sep 2026–May 2027 Machine Learning Internship

High-Dimensional Density Analysis Using Local Unimodality.  
Supervised by Argyris Kalogeratos (Senior Research Scientist) at Centre Borelli, ENS Paris-Saclay, France.

### May–Aug 2026 M1 Research Internship

Optical Modeling of VO<sub>2</sub>-Based Thermochromic Oxide Nanocomposites.  
Supervised by Nguyen Ngoc Duy (Prof.) and Amaury Baret (PhD Student) at the University of Liège, Belgium.

### Apr–May 2026 Research Internship

Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems.  
Supervised by Martin Gander (Prof.) and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

### Jun–Aug 2025 L3 Research Internship

Lattice-Based Post-Quantum Cryptography.  
Supervised by Pierre-Alain Fouque (Prof.) and Aurore Guillevic (Research Scientist) at Inria, France.

### May–Jun 2025 Research Internship

Algebraic Counting of the Real Roots of Polynomials.  
Supervised by Marie-Françoise Roy (Prof. Emeritus) at the University of Rennes, France.

## GitHub Projects

- 2026 **Comparing Dense and Sparse Neural Models for Constrained Graph Generation**  
PyTorch implementation of diffusion-based graph generation with hard connectivity and edge-budget guarantees, with a focus on structural validity and efficient generation.
- 2026 **Optimal Control of Spatial Lotka–Volterra Systems**  
Numerical simulation and constrained optimization of spatial Lotka–Volterra reaction–diffusion systems using an exact discrete adjoint.

## Other Activities

- 2024–2025 **ENS Oral Examinations Preparation**  
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).
- 2022–2025 **Teaching in Mathematics**  
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, and for first- and second-year BSc students at the University of Rennes.
- 2023–2024 **BSc Student Representative**  
Representative on the improvement council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**  
Participation in the Concours Général in Mathematics, Physics, and Philosophy, and the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

## Technical Skills

**Machine Learning:** PyTorch, scikit-learn.

**Scientific Computing:** Python (NumPy, pandas), MATLAB, R.

**Programming:** Python, C/C++, Java.

**Methods:** numerical optimization, optimal control, probabilistic methods.

**Tools:** Git,  $\text{\LaTeX}$ .

## Languages

**French:** native.     **English:** professional working proficiency.

## Sports and Interests

**Sports:** running, cycling, strength training.

**Interests:** fantasy and science fiction literature, chess.

## Academic References

**François Bolley** — [francois.bolley@ens-rennes.fr](mailto:francois.bolley@ens-rennes.fr)

Full Professor, IRMAR, ENS de Rennes.

Director of the Mathematics Department at ENS de Rennes.

**Jean-Christophe Breton** — [jean-christophe.breton@univ-rennes.fr](mailto:jean-christophe.breton@univ-rennes.fr)

Full Professor, IRMAR, University of Rennes.

**Françoise Dal’Bo** — [francoise.dalbo@univ-rennes.fr](mailto:francoise.dalbo@univ-rennes.fr)

Full Professor, IRMAR, University of Rennes.

Director of the International Research Laboratory IFUMI.

**Martin Gander** — [martin.gander@unige.ch](mailto:martin.gander@unige.ch)

Full Professor, Section of Mathematics, University of Geneva.

**Argyris Kalogeratos** — [argyris.kalogeratos@ens-paris-saclay.fr](mailto:argyris.kalogeratos@ens-paris-saclay.fr)

Senior Research Scientist, Centre Borelli, ENS Paris-Saclay.